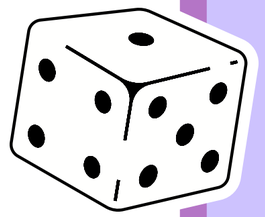
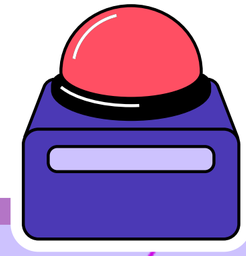
BCC1244 - GAME PROGRAMMING



USER INTERFACE DESIGN

START

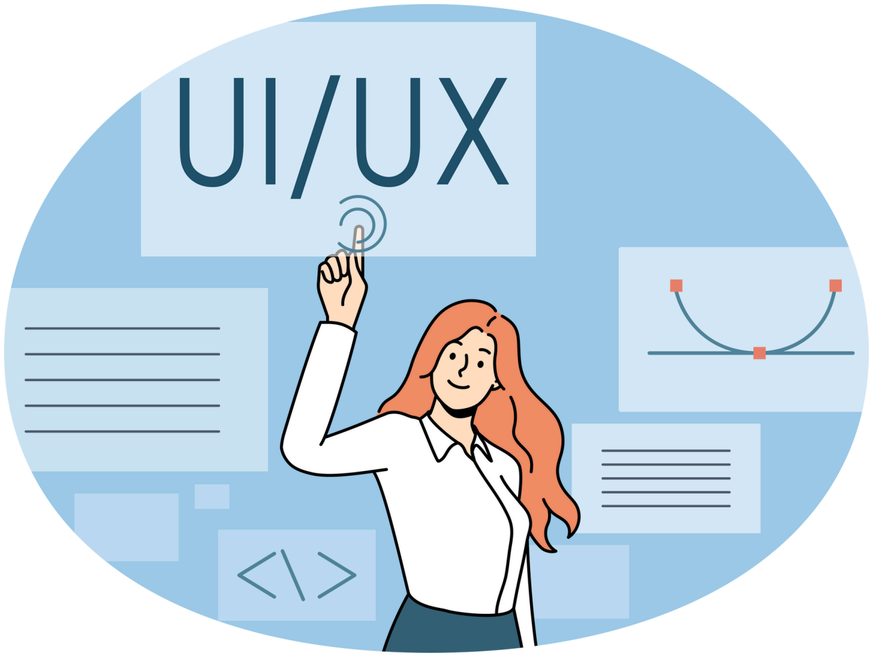


INTRODUCTION TO UI DESIGN

- *The User Interface (UI) in games is the medium through which players interact with the game world, access information, and control gameplay.*
- *Includes HUDs (Heads-Up Displays), menus, icons, buttons, and other interactive elements.*

Importance of UI in Games

- *Enhances player immersion and engagement.*
- *Provides critical feedback and information.*
- *Facilitates navigation and control.*

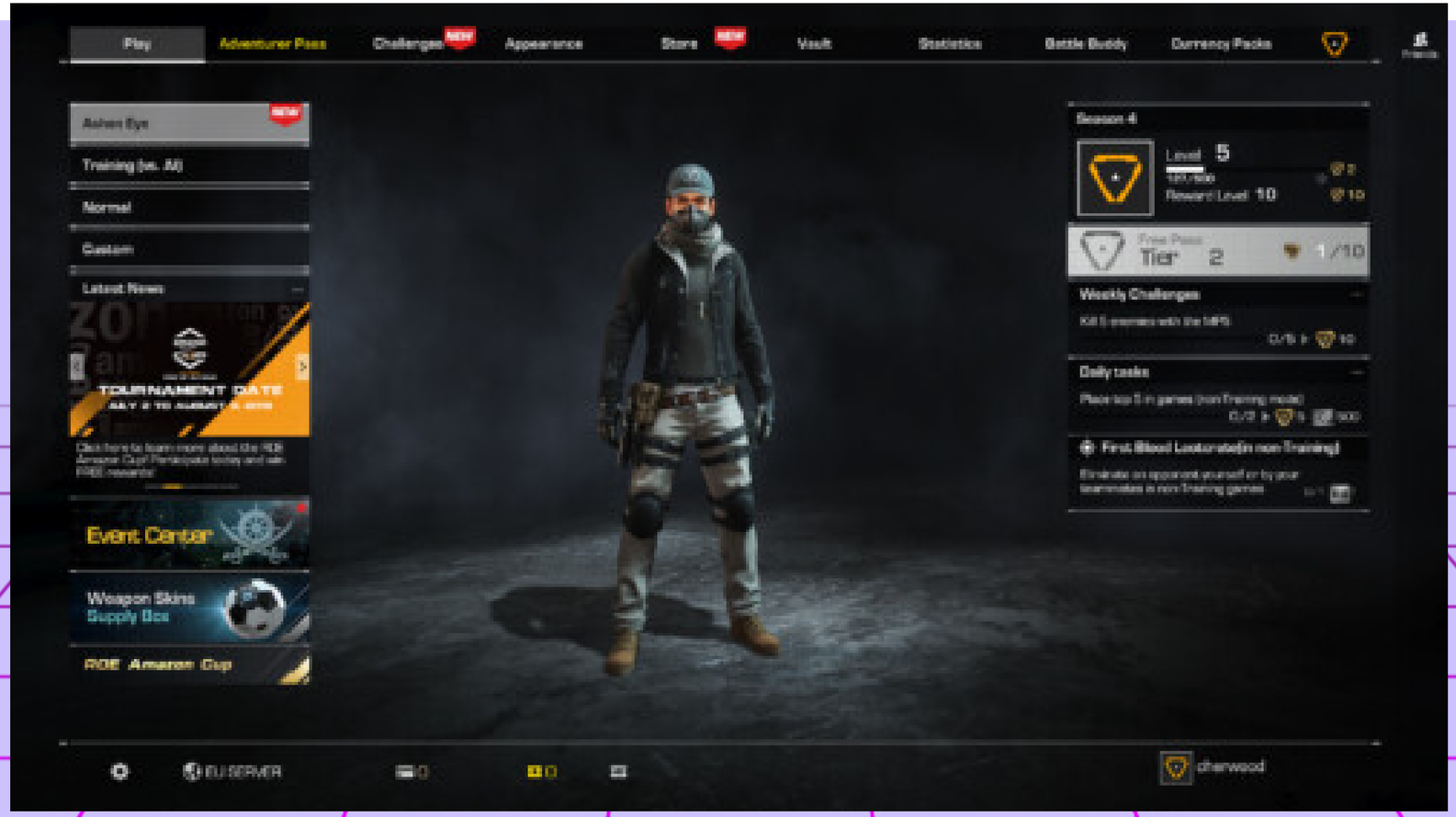


KEY PRINCIPLES OF GAME UI DESIGN

KEY PRINCIPLES

Clarity and Readability

- *Ensure text, icons, and elements are easy to read and understand.*
- *Use appropriate fonts, sizes, and contrast.*



KEY PRINCIPLES

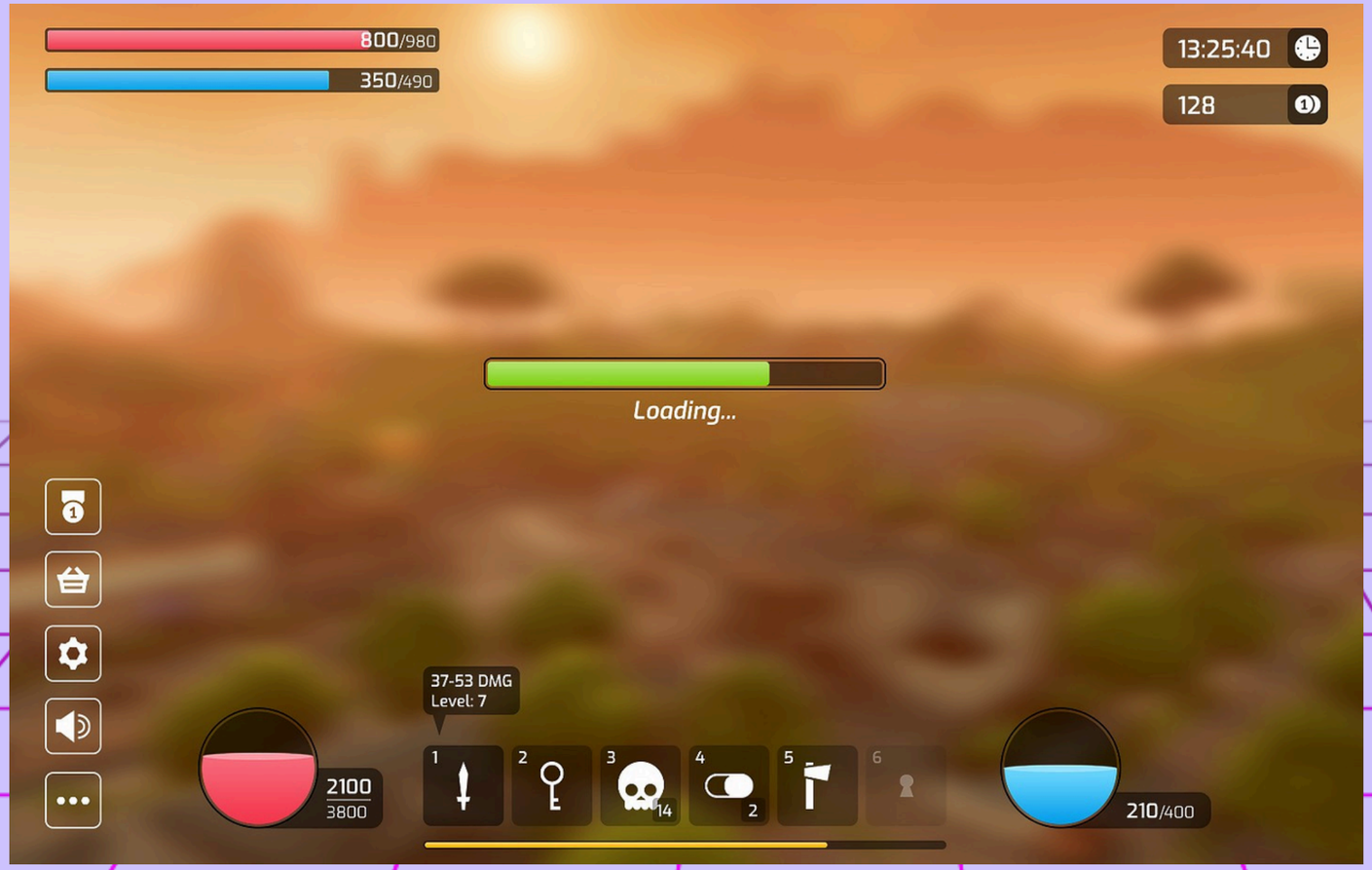
Consistency

- *Maintain a consistent visual style and behavior across all UI elements.*
- *Use familiar conventions (e.g., red for danger, green for health).*

KEY PRINCIPLES

Minimalism

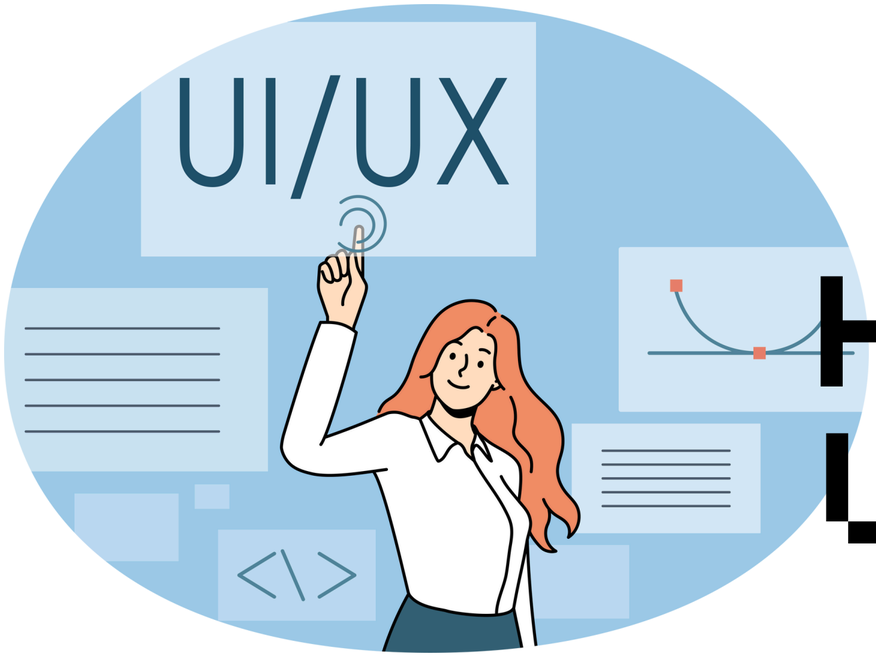
- *Avoid cluttering the screen with unnecessary information.*
- *Prioritize essential elements to reduce cognitive load.*



KEY PRINCIPLES

Accessibility

- *Design for diverse players, including colorblindness, hearing impairments, and motor disabilities.*
- *Offer customizable options (e.g., resizable text, remappable controls).*



DESIGNING HUDS (HEADS-UP DISPLAYS)

DESIGNING HUDS

Purpose of HUDs

- *Display real-time information without interrupting gameplay.*
- *Examples: Health bars, ammo counters, maps, objectives, and timers.*

DESIGNING HUDS

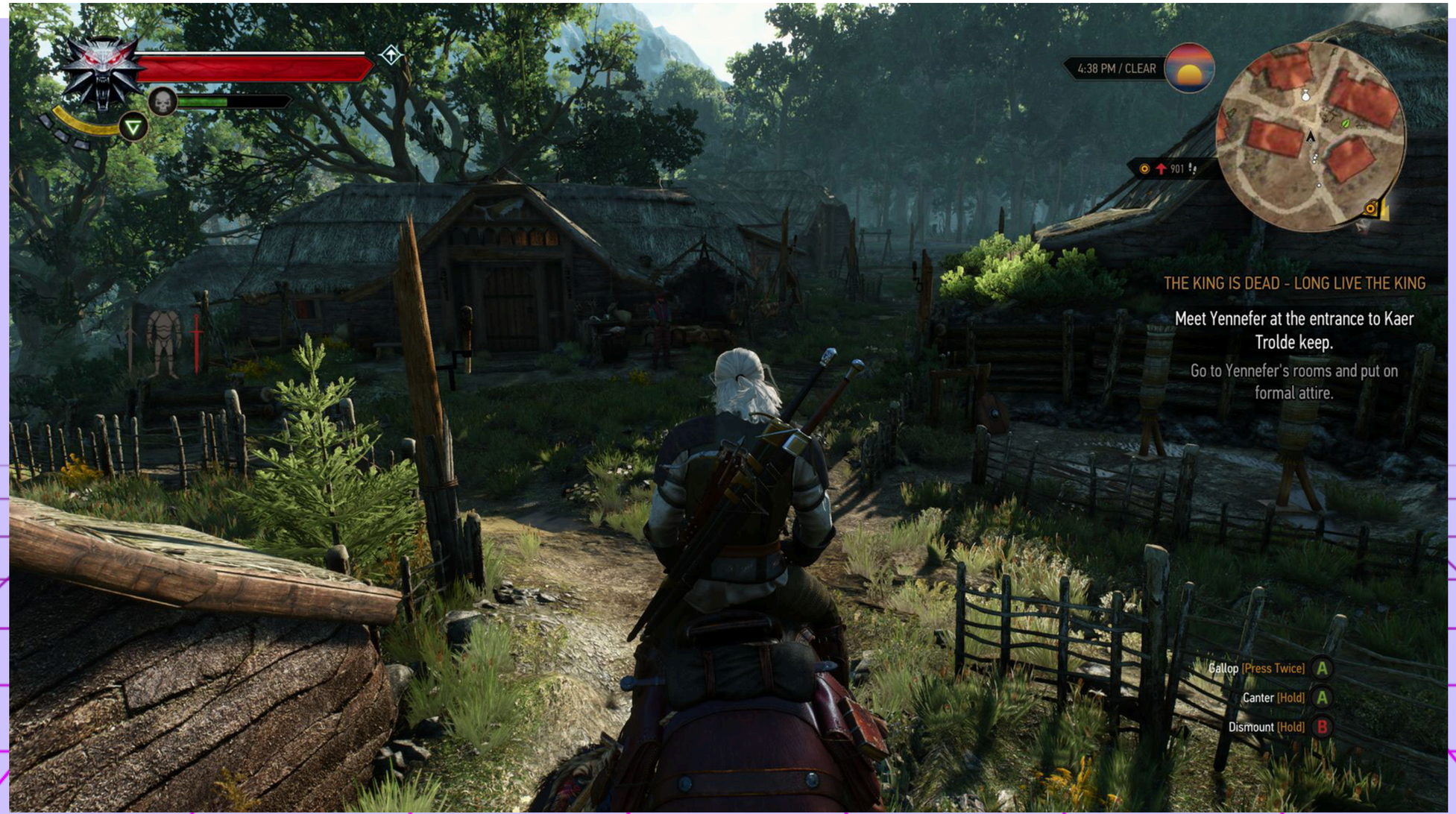
HUD Design Considerations

- *Placement: Position elements where they are easily visible but don't obstruct gameplay (e.g., health bar in the corner, ammo near the crosshair).*
- *Visual Hierarchy: Highlight the most important information (e.g., health > ammo).*
- *Integration: Blend HUD elements with the game's art style (e.g., futuristic HUD for sci-fi games).*

DESIGNING HUDS

Examples of Effective HUDs

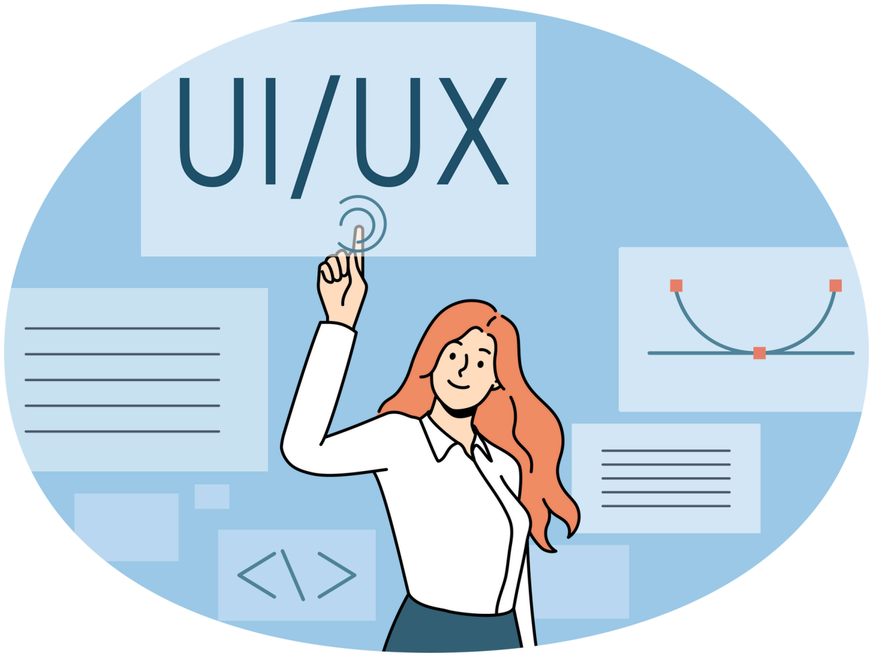
- *The Witcher 3: Minimalist HUD with contextual information.*



DESIGNING HUDS

Examples of Effective HUDs

- *Overwatch: Dynamic HUD that adapts to the player's hero and situation.*



DESIGNING MENUS

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Types of Menus

- *Main Menu: The first screen players see (e.g., Start, Settings, Quit).*
- *Pause Menu: Accessed during gameplay (e.g., Resume, Save, Options).*
- *In-Game Menus: Inventory, maps, or character customization screens.*

DESIGNING MENUS

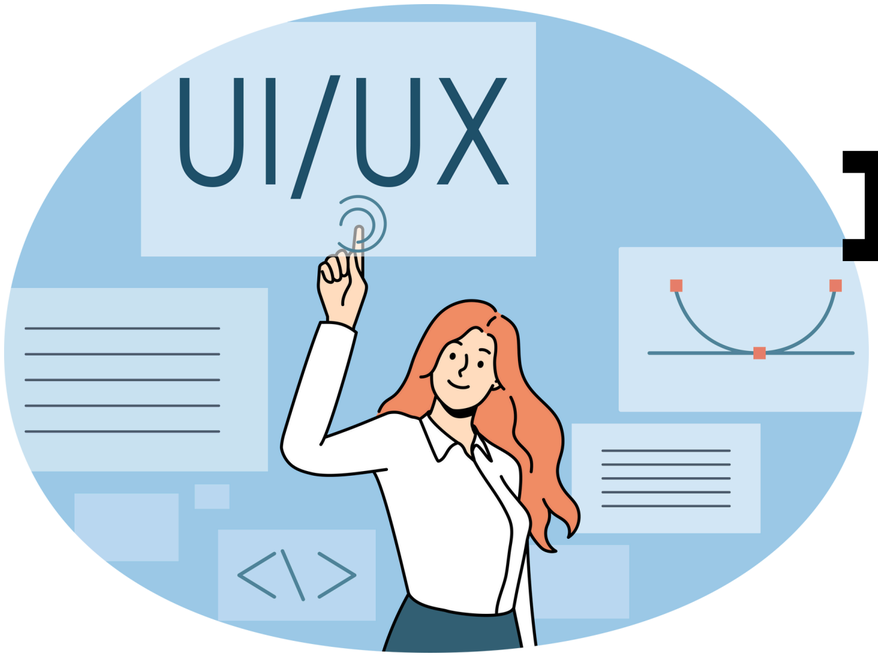
Menu Design Best Practices

- *Navigation: Ensure menus are easy to navigate with clear labels and logical grouping.*
- *Responsiveness: Make menus quick to load and interact with.*
- *Aesthetics: Match the menu design to the game's theme and tone.*

DESIGNING MENUS

Interactive Elements in Menus

- *Buttons, sliders, dropdowns, and toggles.*
- *Use hover effects, animations, and sound to enhance interactivity.*



INTERACTIVE UI ELEMENTS

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Buttons and Icons

- *Design buttons that are visually distinct and easy to click/tap.*
- *Use icons that are universally recognizable (e.g., gear for settings).*

INTERACTIVE UI ELEMENTS

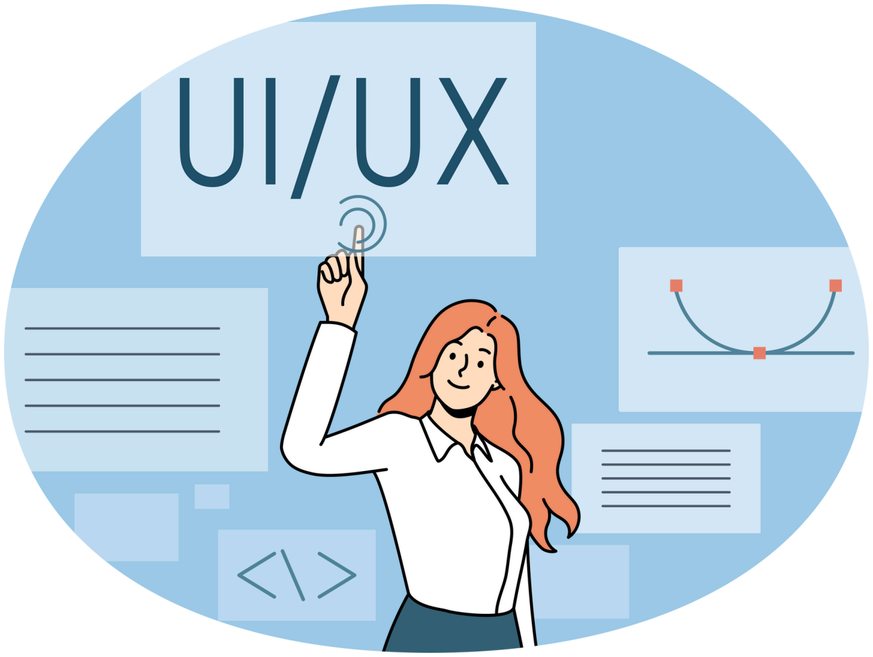
Progress Indicators

- *Show progress through bars, circles, or percentages (e.g., loading screens, experience points).*

INTERACTIVE UI ELEMENTS

Tooltips and Tutorials

- *Provide contextual help through tooltips or guided tutorials.*
- *Keep them concise and optional for experienced players.*



TOOLS AND TECHNIQUES FOR GAME UI DESIGN

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Design Tools

- *Adobe XD, Figma, or Sketch for prototyping.*
- *Unity or Unreal Engine for implementing UI in-game.*

TOOLS AND TECHNIQUES

Implementation Tips

- *Use anchoring and scaling to ensure UI works across different screen sizes.*
- *Optimize for performance to avoid lag or delays in UI responsiveness.*

1

3DS & NDS (side by side), Newer PC/Android

2

Lynx, Wii, PS3, Xbox 360, GBA, PSP, PS Vita, Wii U Switch, PC, Android games

3

NES, SNES, Master System, Genesis, Jaguar, Nintendo 64, GameCube, Dreamcast, PlayStation, PlayStation 2, Xbox, Gameboy, Gameboy Color, Arcade, MS-DOS, Apple (old), Amiga, Legacy PC, Modern indie games

Examples

1

Phone with panoramic controller (Gamesir X2 & G8)

2

Steam Deck, Retroid Pocket 4 Pro

3

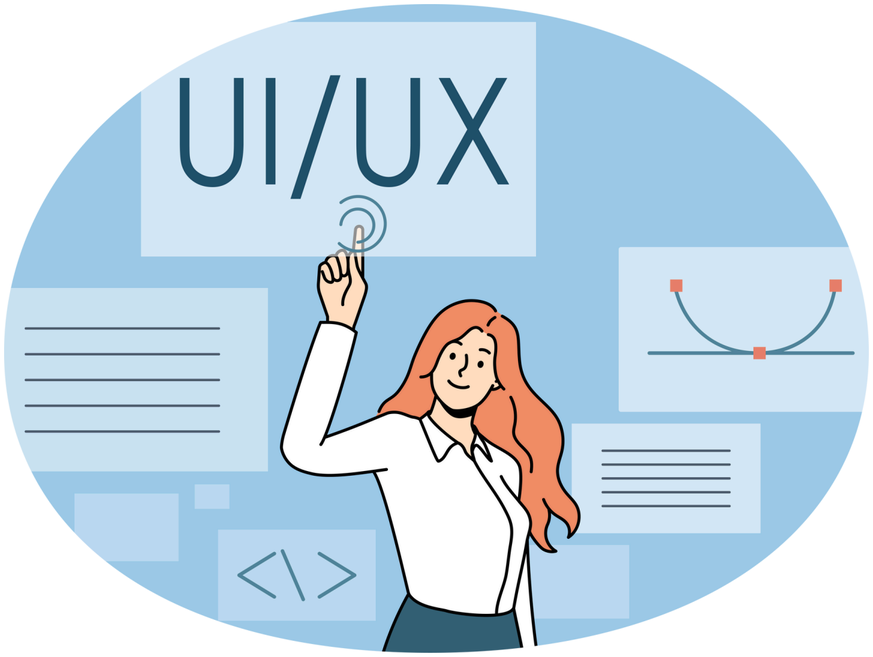
Anbernic RG35XX & RG35XX H

Note: You can run all of them in the device you want as long as it's powerful enough

TOOLS AND TECHNIQUES

Testing and Iteration

- *Conduct playtesting to gather feedback on UI usability.*
- *Iterate based on player behavior and preferences.*



CASE STUDIES

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Case Study 1: Dead Space

- *Innovative HUD integrated into the game world (e.g., health bar on the character's back).*
- *Enhances immersion by minimizing traditional UI elements.*

CASE STUDIES

Case Study 2: Fortnite

- *Clean and colorful UI design that appeals to a broad audience.*
- *Dynamic menus and HUDs that adapt to different game modes.*

THANK YOU FOR PLAYING!